

lab_13

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```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)  
library(DESeq2)
```

Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following object is masked from 'package:dplyr':

explain

The following objects are masked from 'package:base':

as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
setequal, union

Attaching package: 'BiocGenerics'

The following object is masked from 'package:dplyr':

combine

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following objects are masked from 'package:dplyr':

first, rename

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Attaching package: 'IRanges'

The following objects are masked from 'package:dplyr':

collapse, desc, slice

The following object is masked from 'package:grDevices':

windows

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'matrixStats'

The following object is masked from 'package:dplyr':

count

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

```
colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars
```

Loading required package: Biobase

Welcome to Bioconductor

```
Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.
```

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

```
rowMedians
```

The following objects are masked from 'package:matrixStats':

```
anyMissing, rowMedians
```

```
library(AnnotationDbi)
```

Attaching package: 'AnnotationDbi'

The following object is masked from 'package:dplyr':

```
select
```

```
library(org.Hs.eg.db)
```

```
library(pathview)
```

```
#####  
Pathview is an open source software package distributed under GNU General  
Public License version 3 (GPLv3). Details of GPLv3 is available at  
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to  
formally cite the original Pathview paper (not just mention it) in publications  
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)
```

```
metaFile <- "GSE37704_metadata.csv"  
countFile <- "GSE37704_featurecounts.csv"  
  
colData = read.csv(metaFile, row.names=1)  
countData = read.csv(countFile, row.names=1)
```

Q. Complete the code below to remove the troublesome first column from countData

```
countData <- countData[,-1]  
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

```
countData <- countData %>%
  filter(rowSums(across(where(is.numeric))) != 0)
countData <- as.matrix(countData)
```

```
dds = DESeqDataSetFromMatrix(countData=countData,
                              colData=colData,
                              design=~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds = DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```

class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
               ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor

```

```

res = results(dds)
res$logP <- -log10(res$padj)
res$diffexpr <- "No"
res$diffexpr[res$log2FoldChange > 2 & res$padj < 0.05] <- "Up"
res$diffexpr[res$log2FoldChange < -2 & res$padj < 0.05] <- "Down"

```

Q. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

```

out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]     : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results

```

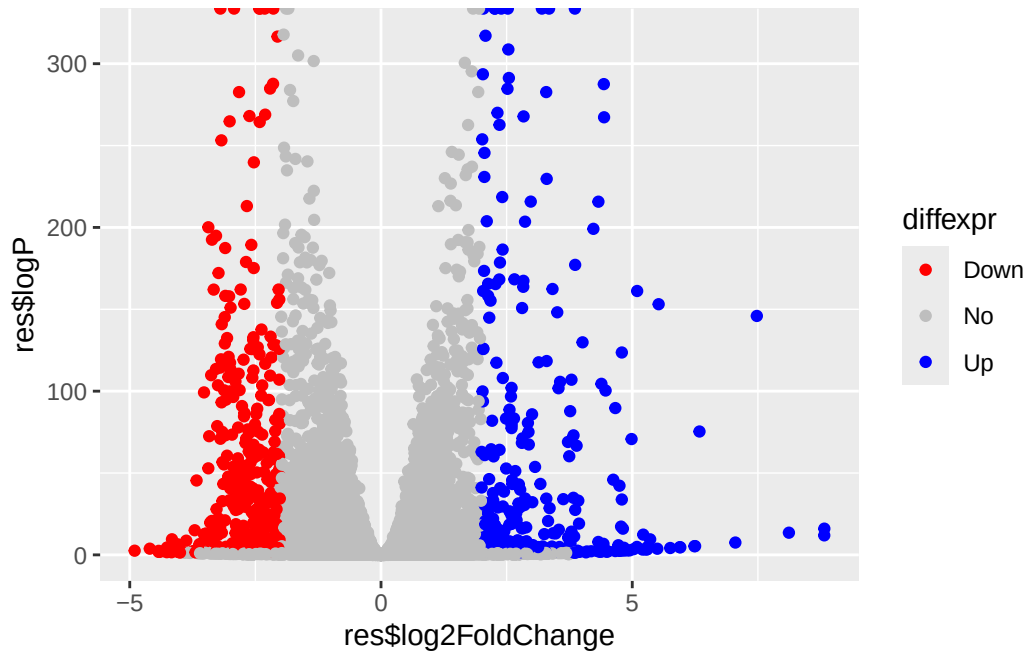
Q. Improve this plot by completing the below code, which adds color and axis labels

```

ggplot(res) +
  aes(res$log2FoldChange, res$logP, col=diffexpr) +
  geom_point() +
  scale_color_manual(values=c("red","gray","blue"))

```

Warning: Removed 1237 rows containing missing values or values outside the scale range (``geom_point()``).



Q. Use the `mapIds()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys=row.names(res), # Our genenames
                     keytype="ENSEMBL",   # The format of our genenames
                     column="SYMBOL",     # The new format we want to add
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys=row.names(res), # Our genenames
                     keytype="ENSEMBL",   # The format of our genenames
                     column="ENTREZID",   # The new format we want to add
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.


```
ord <- order( res$padj )
write.csv(res[ord,], "deseq_results.csv")
```

```
columns(org.Hs.eg.db)
```

```
[1] "ACCNUM"      "ALIAS"      "ENSEMBL"    "ENSEMBLPROT" "ENSEMBLTRANS"
[6] "ENTREZID"    "ENZYME"     "EVIDENCE"   "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"    "GO"         "GOALL"      "IPI"          "MAP"
[16] "OMIM"        "ONTOLOGY"   "ONTOLOGYALL" "PATH"         "PFAM"
[21] "PMID"        "PROSITE"    "REFSEQ"     "SYMBOL"       "UCSCCKG"
[26] "UNIPROT"
```

```
data(kegg.sets.hs)
head(kegg.sets.hs, 2)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"      "1544"    "1548"    "1549"    "1553"    "7498"    "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
[1] "10"      "1066"    "10720"   "10941"   "151531"  "1548"    "1549"    "1551"
[9] "1553"    "1576"    "1577"    "1806"    "1807"    "1890"    "221223"  "2990"
[17] "3251"    "3614"    "3615"    "3704"    "51733"   "54490"   "54575"   "54576"
[25] "54577"   "54578"   "54579"   "54600"   "54657"   "54658"   "54659"   "54963"
[33] "574537"  "64816"   "7083"    "7084"    "7172"    "7363"    "7364"    "7365"
[41] "7366"    "7367"    "7371"    "7372"    "7378"    "7498"    "79799"   "83549"
[49] "8824"    "8833"    "9"       "978"
```

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
<NA>      148398      26155      339451      84069      84808
0.17925708 0.42645712 -0.69272046 0.72975561 0.04057653 0.54281049
```

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/Local_Admin/OneDrive - San Diego State University (SDSU.)

Info: Writing image file hsa04110.pathview.png

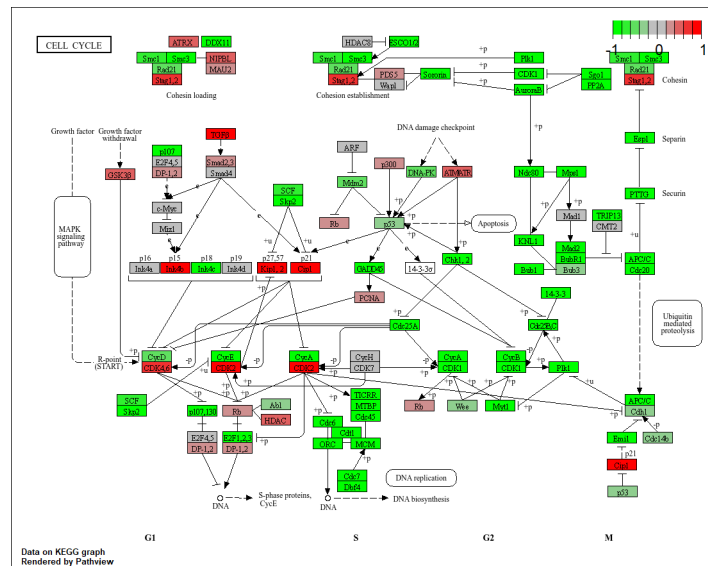


Figure 1: kegg

```
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$greater)[1:5]

# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/Local_Admin/OneDrive - San Diego State University (SDSU.)

Info: Writing image file hsa04060.pathview.png

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory C:/Users/Local_Admin/OneDrive - San Diego State University (SDSU.L
```

```
Info: Writing image file hsa05323.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory C:/Users/Local_Admin/OneDrive - San Diego State University (SDSU.L
```

```
Info: Writing image file hsa05146.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory C:/Users/Local_Admin/OneDrive - San Diego State University (SDSU.L
```

```
Info: Writing image file hsa05332.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory C:/Users/Local_Admin/OneDrive - San Diego State University (SDSU.L
```

```
Info: Writing image file hsa04640.pathview.png
```

```
keggresids
```

```
[1] "hsa04060" "hsa05323" "hsa05146" "hsa05332" "hsa04640"
```

Q. Do the same procedure for the mos 5 downregulated pathways

```
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$less)[1:5]

# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

```
'select()' returned 1:1 mapping between keys and columns
```

Info: Working in directory C:/Users/Local_Admin/OneDrive - San Diego State University (SDSU.L

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/Local_Admin/OneDrive - San Diego State University (SDSU.L

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/Local_Admin/OneDrive - San Diego State University (SDSU.L

Info: Writing image file hsa05130.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/Local_Admin/OneDrive - San Diego State University (SDSU.L

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/Local_Admin/OneDrive - San Diego State University (SDSU.L

Info: Writing image file hsa03440.pathview.png

keggresids

```
[1] "hsa04110" "hsa03030" "hsa05130" "hsa03013" "hsa03440"
```

Same analysis with gene ontology

```

data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets, same.dir=TRUE)

lapply(gobpres, head)

```

\$greater

	p.geomean	stat.mean	p.val
G0:0007156 homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
G0:0002009 morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729 tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610 behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562 epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295 tube development	5.953254e-04	3.253665	5.953254e-04

	q.val	set.size	expl
G0:0007156 homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009 morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729 tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610 behavior	0.1967577	426	1.925222e-04
G0:0060562 epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295 tube development	0.3565320	391	5.953254e-04

\$less

	p.geomean	stat.mean	p.val
G0:0048285 organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280 nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067 mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059 chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236 mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10

	q.val	set.size	expl
G0:0048285 organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280 nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067 mitosis	5.841698e-12	352	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059 chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236 mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

		stat.mean	exp1
G0:0007156	homophilic cell adhesion	3.824205	3.824205
G0:0002009	morphogenesis of an epithelium	3.653886	3.653886
G0:0048729	tissue morphogenesis	3.643242	3.643242
G0:0007610	behavior	3.565432	3.565432
G0:0060562	epithelial tube morphogenesis	3.261376	3.261376
G0:0035295	tube development	3.253665	3.253665

Reactome

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]  
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

Cell cycle. The most significant pathways do not entirely match kegg results. One analysis was done with foldchange and the other with adjusted p-values.